C PROGRAMMING

Lesson 10 — Logical Operators

Based on: Bro Code C Tutorial Series

1. What are Logical Operators?

Logical operators let you combine or modify conditions inside if statements. Instead of writing multiple separate if checks, you can combine them into one clean condition.

C has three logical operators:

- **&&** **(AND)** — both conditions must be true
- **||** **(OR)** — at least one condition must be true
- **!** **(NOT)** — reverses true to false and false to true

2. AND Operator — &&

Both conditions must be true for the whole expression to be true. If even one is false, the result is false.

```
int temp = 20;

if (temp > 0 && temp < 30) {
    printf("Temperature is good\n");
} else {
    printf("Temperature is bad\n");
}
```

3. OR Operator — ||

At least one condition must be true for the whole expression to be true. It only fails if BOTH conditions are false.

```
int temp = 5;

if (temp <= 0 || temp >= 30) {
    printf("Temperature is bad\n");
} else {
    printf("Temperature is good\n");
}
```

4. NOT Operator — !

The NOT operator flips the value of a boolean. true becomes false, and false becomes true. It reads naturally — "if NOT this condition".

```
#include <stdbool.h>

bool isSunny = false;

if (isSunny) {
    printf("It is sunny outside\n");
}

if (!isSunny) {
    printf("It is cloudy outside\n");  // this runs
}
```

5. Combining Logical Operators

You can combine && and || in one condition. Use parentheses to make the logic clear and avoid confusion.

```
int temp = 20;
bool isSunny = true;

// Good weather: temperature in range AND sunny
if ((temp > 0 && temp < 30) && isSunny) {
    printf("Perfect weather!\n");
}

// Bad weather: too hot OR too cold OR not sunny
if (temp <= 0 || temp >= 30 || !isSunny) {
    printf("Stay inside\n");
}
```

6. Nested If vs Logical Operators

Logical operators can often replace nested if statements and make the code cleaner. Here is the Lesson 9 movie ticket example rewritten using &&:

```
// Lesson 9 approach — nested if
if (is_student) {
    if (is_senior) {
        price *= 0.7;
    }
}

// Lesson 10 approach — logical operator
```

```
if (is_student && is_senior) {
    price *= 0.7;   // same result, much cleaner!
}
```

7. Practice Problem

Practice: Ride Eligibility Checker

Write a C program that checks if a person is eligible to ride a roller coaster based on multiple conditions.

Rules:

- Must be at least 12 years old AND at least 120 cm tall to ride
- If under 18, must also have a parent present (bool hasParent)
- Print a clear message explaining why they cannot ride if rejected

Expected Output examples

```
// age=15, height=130, hasParent=true
You can ride!

// age=15, height=130, hasParent=false
Sorry, under 18 requires a parent present.

// age=10, height=130, hasParent=true
Sorry, you must be at least 12 years old.

// age=20, height=110, hasParent=false
Sorry, you must be at least 120cm tall.
```